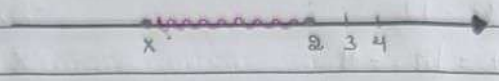
Lista - Intervalos, equações e inequações - Matemática Elementar - MAP191

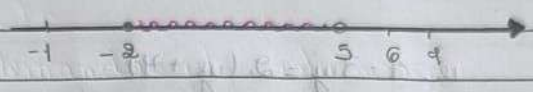
Intervalos

1- Escreva as desigualdades abaixo na forma de intervalos e represente na reta real:

a. $x < 2$ $[x, 2]$



b. $-2 < x < 5$ $[-2, 5[$

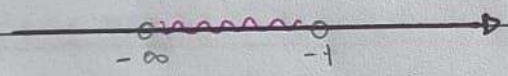


c. $x > 5$ $]x, 5[$

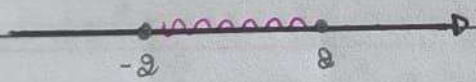
d. $-4 < x < 4$ $]-4, 4[$

2- Escreva os intervalos na forma de desigualdade e represente graficamente na reta real:

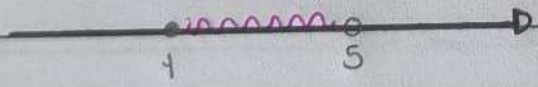
a. $]-\infty, -1[$



b. $[-2, 2]$



c. $[1, 5[$



d. $[3, +\infty[$

Equação

3. Resolva as equações do 1º grau:

a. $3X + 18 = 0$

$3X = 18$

$X = \frac{18}{3} = 6$

b. $3z - 4 = 8$

$3z = 8 + 4$

$3z = 12$

$z = \frac{12}{3} = 4$

c. $2X - 3 = 4X - 5$

$2X - 4X = 3 - 5$

$-2X = -2$

$X = \frac{-2}{-2} = 1$

d. $4(y - 2) = 5y$

$4y - 8 = 5y$

$4y - 5y = 8$

$-1y = 8$

$y = \frac{-8}{-1} = -8$

e. $4 - 3y = 2(y + 4)$

$4 - 3y = 2y + 8$

$-3y - 2y = -4 + 8$

$-5y = 4$

$y = \frac{4}{-5} = -0,8$

f. $\frac{2}{3}X = \frac{4}{5}$

$2X = 3\left(\frac{4}{5}\right)$

$2X = \frac{12}{5}$

$2X = 2,4$

$X = \frac{2,4}{2} = 1,2$

g. $\frac{X}{2} + \frac{1}{3} = 1$

$\frac{3X}{6} + \frac{2}{6} = 1$

$\frac{3X}{6} = 1 - \frac{2}{6}$

$\frac{3X}{6} = \frac{6-2}{6}$

$\frac{3X}{6} = \frac{4}{6}$

$3X = 6\left(\frac{4}{6}\right)$

$3X = \frac{24}{6}$

$3X = 4$

$X = \frac{4}{3} = 1,333$

1b. $\frac{2X-3}{4} + 5 = 3X$

$\frac{2X-3+20}{4} = 3X$

$2X-3+20 = 4(3X)$

$2X-3+20 = 12X$

$2X-12X = 3-20$

$-10X = -19$

$X = \frac{-10}{-19} = 0,5882$

2. Resolva as equações do 2º grau e depois escreva-as na forma fatorada.

1a. $2X^2-3X+1=0$ Logo $X1=1$ e $X2=0,5$

$a=2 \quad b=-3 \quad c=1$

$\Delta = b^2-4ac$

$\Delta = -3^2-4.2.1$

$\Delta = -9-8$

$\Delta = 1$

$X = \frac{-b \pm \sqrt{\Delta}}{2.a} \quad X = \frac{-(-3) \pm \sqrt{1}}{2.2} \quad X = \frac{3 \pm 1}{4}$

$X' = \frac{3+1}{4} = \frac{4}{4} = 1$

$X'' = \frac{3-1}{4} = \frac{2}{4} = 0,5$

Forma fatorada $= 2X^2-3X+1 = 2(X-1)(X-0,5)$

$2(X^2-0,5X-X+0,5)$

$2X^2-X-2X+1$

$2X^2-3X+1$

1b. $4x^2 - 8x + 3 = 0$ Logo $x1 = 1,5$ e $x2 = 0,5$

$a = 4$ $b = -8$ $c = 3$

$\Delta = b^2 - 4ac$

$\Delta = -8^2 - 4.4.3$

$\Delta = 64 - 48$

$\Delta = 16$

$x = \frac{-b \pm \sqrt{\Delta}}{2a}$ $x = \frac{-(-8) \pm \sqrt{16}}{2.4} = \frac{8 \pm 4}{8}$

$x' = \frac{8+4}{8} = \frac{12}{8} = 1,5$

$x'' = \frac{8-4}{8} = \frac{4}{8} = 0,5$

Forma fatorada $= 4x^2 - 8x + 3 = 4(x - 1,5)(x - 0,5)$

$4(x^2 - 0,5x - 1,5x + 0,75)$

$4x^2 - 2x - 6x + 3$

$4x^2 - 8x + 3$

1c. $x^2 - x - 20 = 0$ Logo $x1 = 5$ e $x2 = 4$

$a = 1$ $b = -1$ $c = -20$

$\Delta = b^2 - 4ac$

$\Delta = -1^2 - 4.1 - 20$

$\Delta = 1 + 80$

$\Delta = 81$

$x = \frac{-b \pm \sqrt{\Delta}}{2a}$ $x = \frac{-(-1) \pm \sqrt{81}}{2.1} = \frac{1 \pm 9}{2}$

$x' = \frac{1+9}{2} = \frac{10}{2} = 5$

$x'' = \frac{1-9}{2} = \frac{8}{2} = 4$

Forma fatorada $= x^2 - x - 20 = 1(x - 5)(x - 4)$

$x^2 + 4x - 5x - 20$

$x^2 - x - 20$

1d. $x^2 - 2x + 1 = 0$ Logo $x = 1$

$a = 1$ $b = -2$ $c = +1$

$\Delta = b^2 - 4ac$

$\Delta = -2^2 - 4.1 + 1$

$\Delta = 4 - 4$

$\Delta = 0$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-2) \pm \sqrt{0}}{2.1} = \frac{2}{2} = 1$

Forma fatorada $= x^2 - 2x + 1 = 1(x-1)(x-1)$

$x^2 - x - x + 1$

$x^2 - 2x + 1$

1e. $x^2 - 6x + 9 = 0$ Logo $x = 3$

$a = 1$ $b = -6$ $c = 9$

$\Delta = b^2 - 4ac$

$\Delta = -6^2 - 4.1.9$

$\Delta = 36 - 36$

$\Delta = 0$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-6) \pm \sqrt{0}}{2.1} = \frac{6}{2} = 3$

Forma fatorada $= x^2 - 6x + 9 = 1(x-3)(x-3)$

$x^2 - 3x - 3x + 9$

$x^2 - 3x + 9$

f. $2x^2 - 3x = 0$ Logo $x1 = 1,5$ e $x2 = 0$

$a = 2$ $b = -3$ $c = 0$

$\Delta = b^2 - 4ac$

$\Delta = -3^2 - 4.2.0$

$\Delta = 9 - 0$

$\Delta = 9$

$x = \frac{-b \pm \sqrt{\Delta}}{2.a} = \frac{-(-3) \pm \sqrt{9}}{2.2} = \frac{3 \pm 3}{4}$

$x' = \frac{3+3}{4} = \frac{6}{4} = 1,5$

$x'' = \frac{3-3}{4} = \frac{0}{4} = 0$

Forma fatorada $= 2x^2 - 3x = 2(x - 1,5)(x - 0)$

$2(x^2 - 0 - 1,5x - 0)$

$2x^2 - 3x$

g. $x^2 - 8x = -15$ Logo $x1 = 5$ e $x2 = 3$

$a = 1$ $b = -8$ $c = 15$

$\Delta = b^2 - 4ac$

$\Delta = -8^2 - 4.1.15$

$\Delta = 64 - 60$

$\Delta = 4$

$x = \frac{-b \pm \sqrt{\Delta}}{2.a} = \frac{-(-8) \pm \sqrt{4}}{2.1} = \frac{8 \pm 2}{2}$

$x' = \frac{8+2}{2} = \frac{10}{2} = 5$

$x'' = \frac{8-2}{2} = \frac{6}{2} = 3$

Forma fatorada $= x^2 - 8x + 15 = 1(x - 5)(x - 3)$

$x^2 - 3x - 5x + 15$

$x^2 - 8x + 15$

i- $x^2 = 3x + 4$ Logo $x1 = 4$ e $x2 = -1$

$a = 1$ $b = -3$ $c = -4$

$\Delta = b^2 - 4ac$

$\Delta = -3^2 - 4.1.-4$

$\Delta = 9 + 16$

$\Delta = 25$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-3) \pm \sqrt{25}}{2.1} = \frac{3 \pm 5}{2}$

$\frac{3+5}{2} = \frac{8}{2} = 4$

$\frac{3-5}{2} = \frac{-2}{2} = -1$

Forma fatorada $= x^2 - 3x - 4 = 1(x-4)(x-(-1))$

$x^2 + x - 4x - 4$

$x^2 - 3x - 4$

ii- $x^2 - 49 = 0$ Logo $x1 = 7$ e $x2 = -7$

$a = 1$ $b = 0$ $c = -49$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4.1(-49)$

$\Delta = 0 + 196$

$\Delta = 196$

$x = \frac{-b \pm \sqrt{\Delta}}{2.a} = \frac{-0 \pm \sqrt{196}}{2.1} = \frac{0 \pm 14}{2}$

$\frac{0+14}{2} = \frac{14}{2} = 7$

$\frac{0-14}{2} = \frac{-14}{2} = -7$

Forma fatorada $= x^2 - 49 = 1(x-7)(x-(-7))$

$x^2 + \cancel{7x} - \cancel{7x} - 49$

$x^2 - 49$

j- $4x^2 - 9 = 0$ Logo $x1 = 1,5$ e $x2 = -1,5$

$a = 4$ $b = 0$ $c = -9$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 . 4(-9)$

$\Delta = 0 + 144$

$\Delta = 144$

$x = \frac{-b \pm \sqrt{\Delta}}{2.a} = \frac{-0 \pm \sqrt{144}}{2.4} = \frac{-0 \pm 12}{8}$

$x' = \frac{0+12}{8} = \frac{12}{8} = 1,5$

$x'' = \frac{0-12}{8} = \frac{-12}{8} = -1,5$

Forma Fatorada $= 4x^2 - 9 = 4(x - 1,5)(x - (-1,5))$

$4(x^2 + \cancel{1,5} - \cancel{1,5}x - 2,25)$

$4x^2 - 9$

k- $x^2 + 3x = 0$ Logo $x1 = 0$ e $x2 = -3$

$a = 1$ $b = 3$ $c = 0$

$\Delta = b^2 - 4ac$

$\Delta = 3^2 - 4 . 1 . 0$

$\Delta = 9 - 0$

$\Delta = 9$

$x = \frac{-b \pm \sqrt{\Delta}}{2.a} = \frac{-3 \pm \sqrt{9}}{2.1} = \frac{-3 \pm 3}{2}$

$x' = \frac{-3+3}{2} = \frac{0}{2} = 0$

$x'' = \frac{-3-3}{2} = \frac{-6}{2} = -3$

Forma Fatorada $= x^2 + 3x = 1(x - 0)(x - (-3))$

$x^2 + 3x - \cancel{0} + \cancel{0})$

$x^2 + 3x$

Jandaia

Inequações

5- Resolva as inequações:

1a. $5x+4 < -3 =$ (Menor)

$5x+4 < -12$

$5x < -12-4$

$5x < -16$

$x < \frac{-16}{5}$

$x < -3,8$

1b. $3x+3 < x+6 = 3x-x < 3+6$

$2x < 3$

$x = \frac{3}{2}$

$x = 1,5$

1c. $2x-1 \geq 5x+3 = 2x-5x \geq 3+1$

$-3x \geq 4 \quad \cdot(-1)$

$3x \leq -4$

$x \leq \frac{-4}{3}$

$x \leq -1,333$

1d. $x+3 < 6x-2$

$x-6x < -2-3$

$-5x < -5 \quad \cdot(-1)$

$5x > 5$

$x \geq \frac{5}{5}$

$x \geq 1$

1e. $2x+1 \geq 3x$

$2x-3x \geq -1$

$-x \geq -1 \quad \cdot(-1)$

$x \leq 1$

1f. $2x+5 < 3x-7$

$2x-3x < -7-5$

$-x < -12 \quad \cdot(-1)$

$x > 12$

1g. $3x-7 \geq 5$

$3x \geq 5+7$

$3x \geq 12$

$x \geq \frac{12}{3}$

$x \geq 4$

1h. $2 < x+6 < 9 =$

$2-6 < x < 9-6$

$-4 < x < 3$

1i. $3 < 2x-3 < 7 = 15 < 2x-3 < 35$ (5)

$15+3 < 2x < 35+3$

$18 < 2x < 38$

$\frac{18}{2} < x < \frac{38}{2}$

$9 < x < 19$